HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **monitoring, alerting, and observability solutions**, creating reusable, code-defined dashboards for critical database services.
- Automated system and component recovery using **Infrastructure-as-Code** (CloudFormation) and configuration management to reduce manual effort in production environments.
- Participated in on-call rotation, acting as a senior escalation point for the recovery of major **production incidents** and performing root cause analysis.
- Created **monitoring queries** and established service-level baselines for critical database services, ensuring adherence to defined performance metrics.
- Planned and executed **failover testing** within and between AWS regions to validate disaster recovery strategies and ensure service resilience.
- Developed **SQL scripts** for database design, optimization, and troubleshooting, including schema design, query tuning, and indexing strategies to resolve performance bottlenecks.
- Wrote **Python scripts** to automate operational tasks and data processing, reducing manual intervention in system maintenance and data handling workflows.

SKILLS

Cloud Database Services – Aurora MySQL, Amazon RDS, Amazon Redshift, Amazon DynamoDB, Amazon ElastiCache

Database Skills – SQL, Database Design, Query Optimization, Indexing, Locks, Performance Bottlenecks, Schema Design

Cloud Platforms & IaC – AWS, CloudFormation, Terraform

Automation & Scripting – Python, Bash

Observability & Monitoring – Monitoring, Alerting, Dashboards, Incident Response, Root Cause Analysis

DevOps & CI/CD – CI/CD practices, Database Change Management

PROJECTS

AWS Database Reliability Engineering Platform

- Engineered a platform for managing and monitoring AWS database services, focusing on high availability and fault tolerance.
- Implemented automated recovery scripts in **Python** and **CloudFormation** to handle common database failure scenarios, reducing manual recovery time by an estimated 40%.
- Developed comprehensive **monitoring and alerting** dashboards using SQL queries against database logs and performance metrics.

SQL Performance Optimization and Automation

- Analyzed and refactored inefficient **SQL queries** across enterprise systems to improve performance and reduce resource consumption.
- Developed **Python scripts** to automate database health checks and generate performance reports for key operational databases, covering over 50 critical tables.
- Implemented strategic **indexing** and schema adjustments that decreased average query execution time for key reporting functions by 30%.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)